

1.INTRODUCTION

Automation has become an essential part of modern life, reducing human effort and increasing efficiency in daily activities. Robotics technology is widely used in industries, healthcare, homes, and commercial spaces to perform repetitive and time-consuming tasks. One such important application of automation is floor cleaning, which requires continuous manual effort and consumes significant time.

The “Bluetooth Controlled Manual and Automatic Floor Cleaning Robot” is designed to automate the process of floor cleaning using embedded systems and sensor technology. The system is developed using Arduino Uno as the main controller, which manages the movement and cleaning operations of the robot. The robot operates in two modes: manual mode and automatic mode. In manual mode, the robot is controlled wirelessly using a HC-05 Bluetooth Module through a mobile application. In automatic mode, the robot uses an HC-SR04 Ultrasonic Sensor to detect obstacles and navigate accordingly.

The robot is equipped with DC motors for movement, a water control mechanism for mopping, and an LCD display to show system status. This project aims to provide a low-cost, efficient, and user-friendly cleaning solution suitable for homes, hospitals, colleges, and offices. By integrating hardware and software technologies, the system reduces human intervention while improving cleaning efficiency.

1.1 MOTIVATION

The motivation for this project comes from the daily difficulty of manual floor cleaning, which is time-consuming and tiring. As students, we noticed the need for a simple and affordable solution to make cleaning easier.

This project is motivated by:

- Time-saving requirement for daily tasks
- Difficulty in maintaining regular cleanliness
- Need for a low-cost and efficient solution
- Desire to develop a practical real-life application

1.2 PROBLEM IDENTIFICATION

Floor cleaning is a time-consuming and tiring task, especially for busy students and working individuals. Manual methods require continuous effort and are not always efficient, while existing automatic cleaners are expensive. Therefore, we are working on a low-cost, Bluetooth-controlled automatic floor cleaning robot to reduce human effort and improve cleaning efficiency.

- Manual floor cleaning is time-consuming and requires continuous effort
- Difficult for busy people to maintain daily cleaning
- Traditional methods are not efficient in reaching all areas
- Cleaning can be physically tiring and uncomfortable
- Need for wireless control for ease of operation
- Need to reduce human effort and save time

1.3 OBJECTIVES OF STUDY

The main objectives of this project are:

- To reduce human effort in daily cleaning tasks
- To save time by automating the cleaning process
- To implement Bluetooth control for easy operation
- To integrate ultrasonic sensor for obstacle detection
- To improve cleaning efficiency using a mopping mechanism
- To ensure user-friendly and convenient operation

1.4 SCOPE OF STUDY

The scope of this project includes:

- Design and development of a Bluetooth-controlled cleaning robot
- Implementation of obstacle detection system
- Automation of floor cleaning process
- Development of a low-cost and efficient system
- Application for indoor cleaning (homes and small areas)

2. LITERATURE SURVEY

The development of floor cleaning robots has gained significant attention due to the increasing need for automation in daily life. Various researchers have proposed different types of cleaning robots based on manual control, autonomous operation, and hybrid systems.

Gorade et al. (2022) designed a Bluetooth-controlled floor cleaning robot using an Arduino microcontroller, motor driver, and DC motors. The robot is operated through a smartphone using Bluetooth communication. It includes a simple cleaning mechanism consisting of a water sprayer and mop. The system is cost-effective and easy to implement; however, it depends entirely on user control and lacks autonomous decision-making capabilities.

Kuldeep Soni (2017) proposed an autonomous floor cleaning robot that operates without human intervention. The system uses sensors such as infrared and ultrasonic sensors to detect obstacles and navigate the environment. The robot follows a predefined path or algorithm to clean the floor. While this reduces human effort, the system is more complex and costly due to sensor integration and programming requirements.

Choudhari et al. (2024) developed a smart floor cleaning robot combining Bluetooth control with autonomous navigation. The system uses sensors for obstacle detection and a mobile interface for manual control. This approach provides both flexibility and automation. However, challenges remain in terms of cost, power consumption, and advanced navigation techniques.

From the above studies, it is observed that bluetooth-controlled floor cleaning robots focus on improving efficiency, flexibility, and automation. The integration of Bluetooth control, sensor-based navigation, and multifunctional cleaning mechanisms has significantly enhanced the performance and usability of these systems.

3.BLOCK DIAGRAM

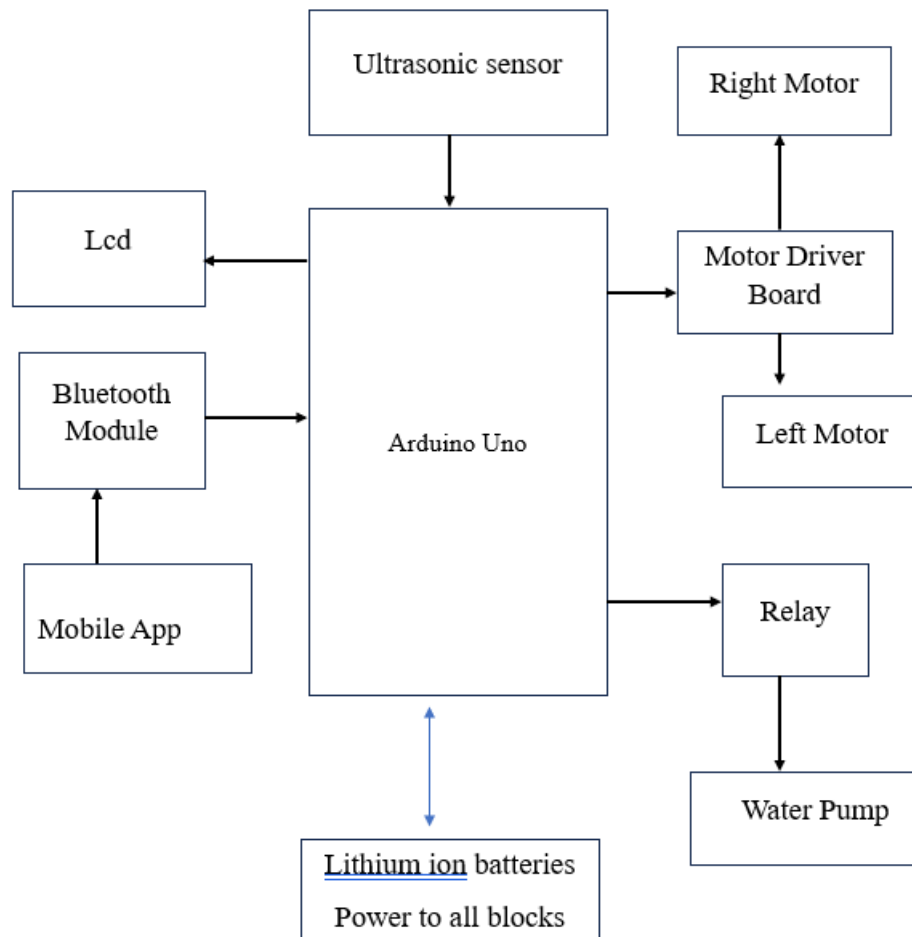


Fig 3.1 Block Diagram

3.2 DESCRIPTION OF BLOCK DIAGRAM

1) Arduino Uno (Main Controller)

Acts as the brain of the system.

Controls all components and processes input/output signals.

2) Bluetooth Module (HC-05)

Enables wireless communication with a smartphone.

Used for manual control of the robot (forward, backward, left, right).

3) Ultrasonic Sensors (Front, Left, Right)

Detect obstacles in different directions.

Help in automatic mode to avoid collisions.

4) Motor Driver

Receives signals from Arduino.

Controls the movement of DC motors (speed and direction).

5) DC Motors

Responsible for movement of the robot.

Enable forward, backward, and turning actions.

6) Relay Module

Works as a switch controlled by Arduino.

Turns the water pump ON/OFF.

7) Water Pump

Sprays water for floor cleaning.

Activated during cleaning process.

8) LCD Display

Displays system status (mode, direction, etc.).

Helps user understand robot operation.

9) Voltage Regulator

Maintains stable voltage supply.

Protects components from damage.

10) Li-ion Batteries

Provide power to the entire system.

Portable energy source for the robot.

3.3 FLOWCHART

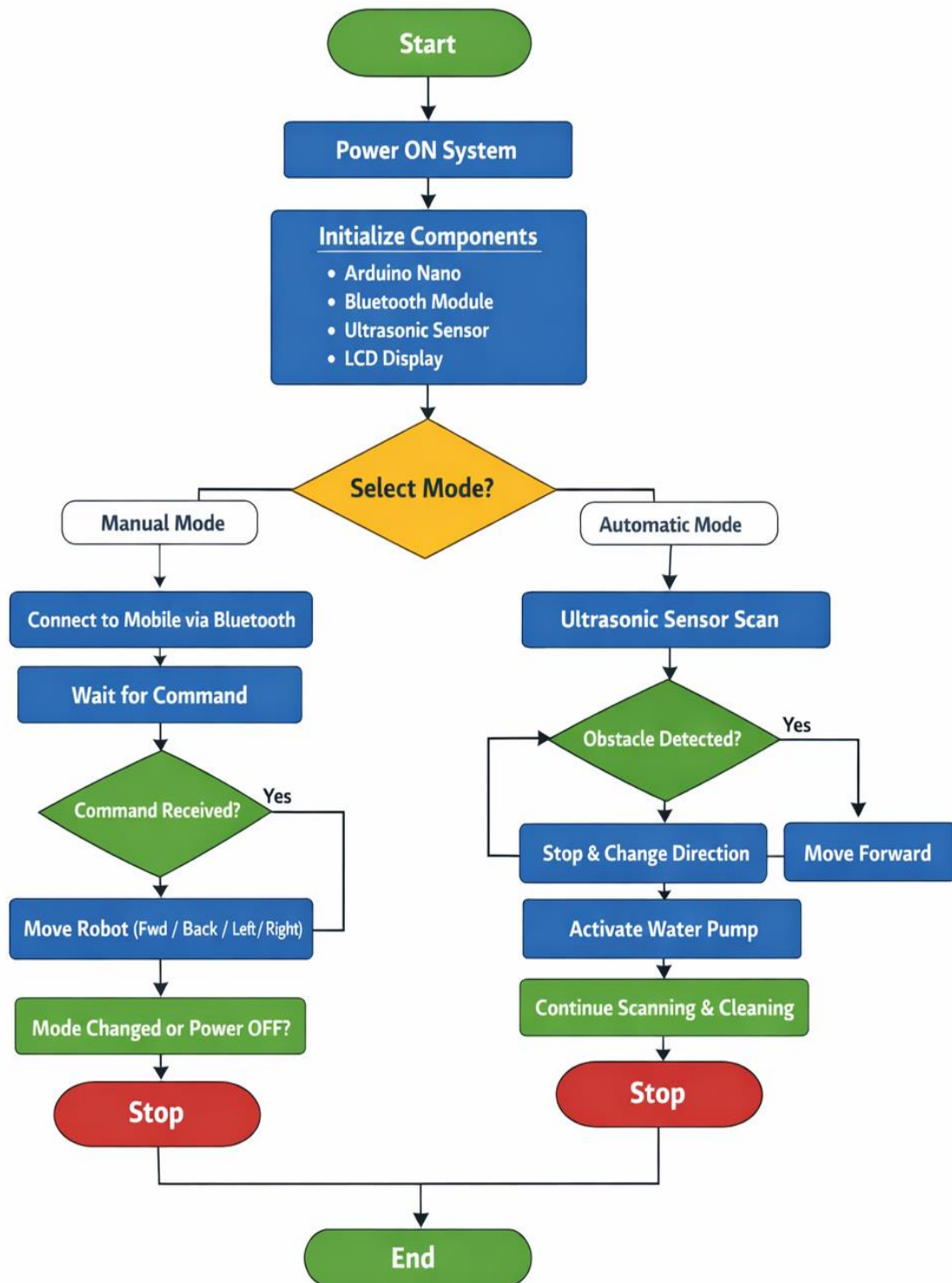


Fig 3.3-Flow chart

3.4 CIRCUIT DIAGRAM

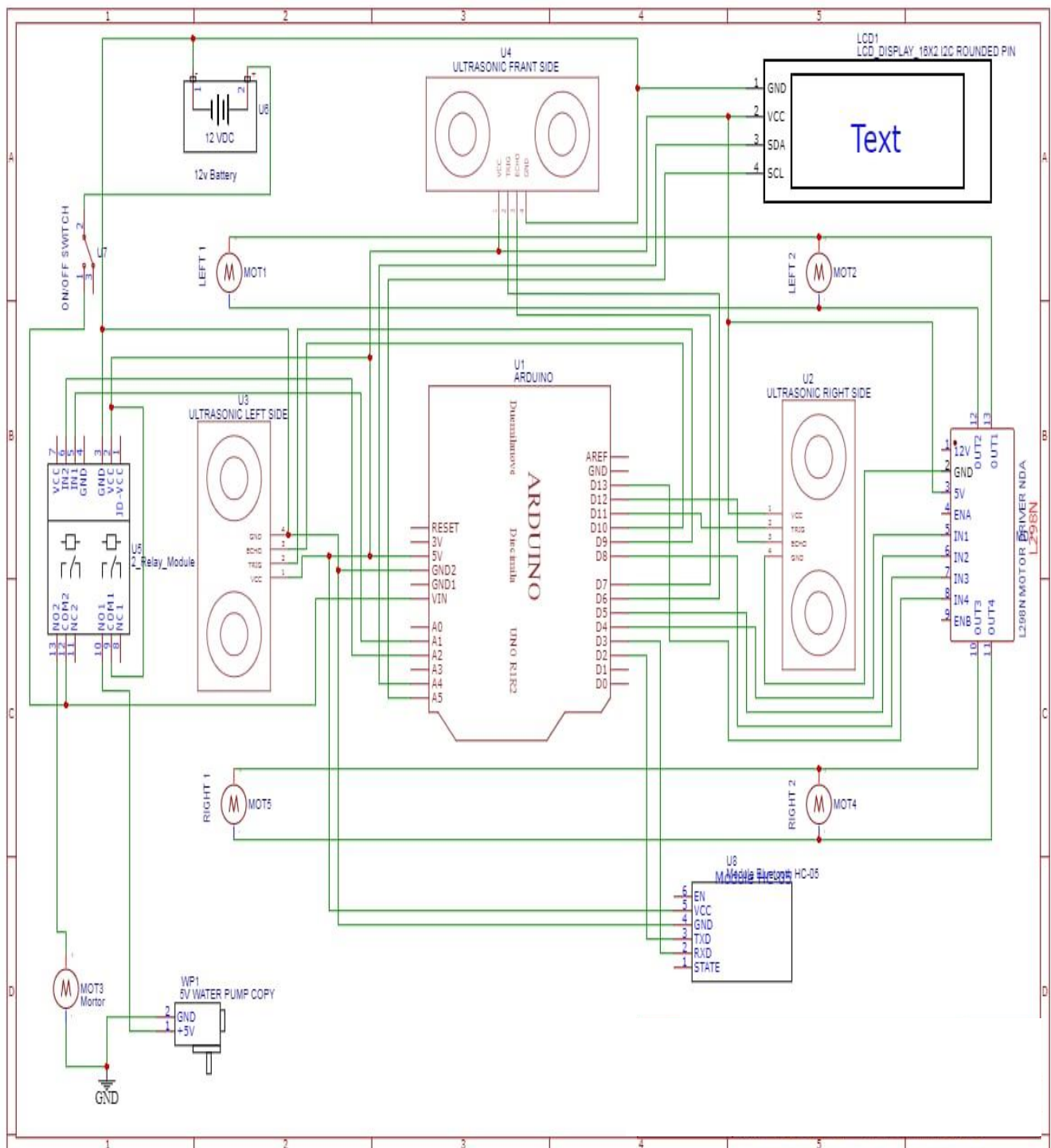


Fig 3.4 Circuit diagram

4.SYSTEM DEVELOPMENT

The system is developed by integrating hardware components like sensors, actuators, and control units using the Arduino Nano as the main controller. The robot operates through programmed instructions, where inputs from sensors are processed and appropriate outputs are generated. Wireless communication via Bluetooth enables user control, while the cleaning mechanism is achieved using motors and a water pump.

HARDWARE REQUIREMENT

- Arduino Nano – Main controller of the robot
- HC-05 Bluetooth Module – For wireless communication
- HC-SR04 Ultrasonic Sensor – For obstacle detection
- DC Motors – For robot movement
- Motor Driver Module / Relay – To control motors
- Lithium-ion Rechargeable Batteries – Power supply
- 16x2 LCD Display – To display system status
- I2C
- Water Pump – For mopping mechanism
- Voltage Regulator – To maintain constant voltage
- Wheels and Chassis
- Connecting Wires
- PCB / Breadboard
- Resistors
- Wires
- Relay
- Male and Female Headers
- Nuts and Bolts
- Glue Gun and Soldering Gun

SOFTWARE REQUIREMENT

- Arduino IDE – For coding and uploading program
- Windows operating system

4.COMPONENT DESCRIPTION

4.1 Arduino Uno

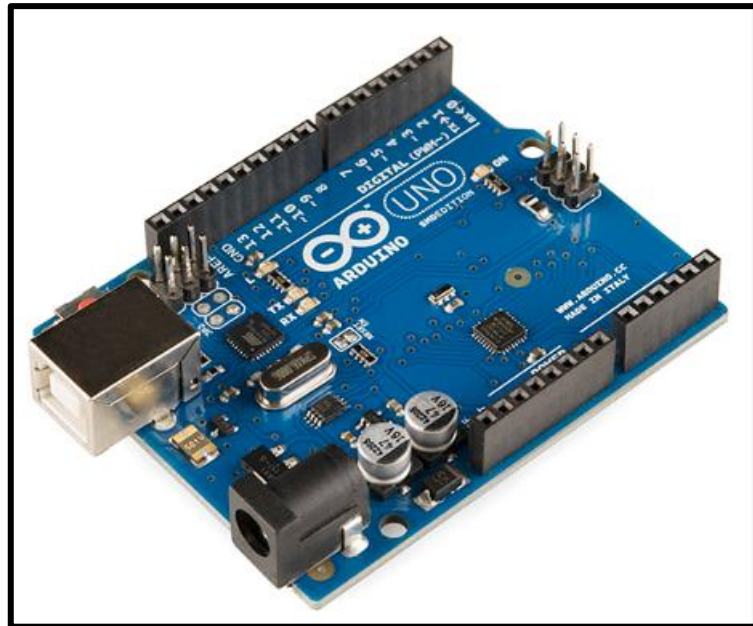


Fig.4.1 Arduino Uno

The **Arduino Uno** is the main controller of the robot. It is a widely used and beginner-friendly microcontroller board based on the **ATmega328P**, responsible for processing input signals from sensors and controlling outputs such as motors, relay modules, LCD display, and water pump.

Due to its robust design, easy availability, and seamless integration with the **Arduino IDE**, it is one of the most popular boards in embedded systems and robotics projects. It supports real-time control and efficient multitasking, making it ideal for automation applications. Additionally, it can be powered through USB or an external power supply, offering flexibility in design and implementation.

Key Features

Microcontroller: ATmega328P

- Operating Voltage: 5V
- Input Voltage: 7–12V
- Digital I/O Pins: 14 (6 PWM)
- Analog Input Pins: 6

- Communication: UART, I2C, SPI
- USB Interface: Type-B USB (for programming)

4.2 HC-05 Bluetooth module

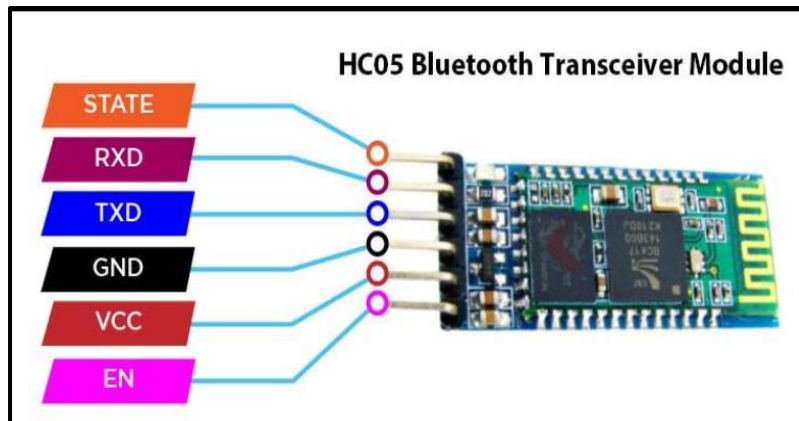


Fig:4.2 HC-05 Bluetooth module

The HC-05 Bluetooth module enables wireless communication between the robot and a smartphone, allowing the user to control its movement and cleaning functions through a mobile application. It operates using Bluetooth technology and communicates with the Arduino Nano via serial (UART) communication.

The module receives commands such as forward, backward, left, right, and cleaning control, making manual operation simple and efficient. It can be easily configured using AT commands for setting parameters like name, password, and baud rate. Due to its reliable performance, low cost, and easy interfacing, the HC-05 is widely used in robotics and automation projects for short-range wireless control.

The TX (Transmit) pin of the HC-05 is used to send data from the Bluetooth module to the Arduino. This pin is connected to the RX pin of the Arduino Nano, allowing the Arduino to receive commands such as movement directions or cleaning controls sent from the mobile application.

The RX (Receive) pin of the HC-05 is used to receive data from the Arduino. It is connected to the TX pin of the Arduino Nano, enabling the Arduino to send responses or control signals back to the module if required.

◆ Key Features

- Frequency: 2.4 GHz
- Operating Voltage: 3.3V (accepts 5V via onboard regulator)
- Range: ~10 meters
- Communication: Serial (UART)

4.3 Motor Driver Module

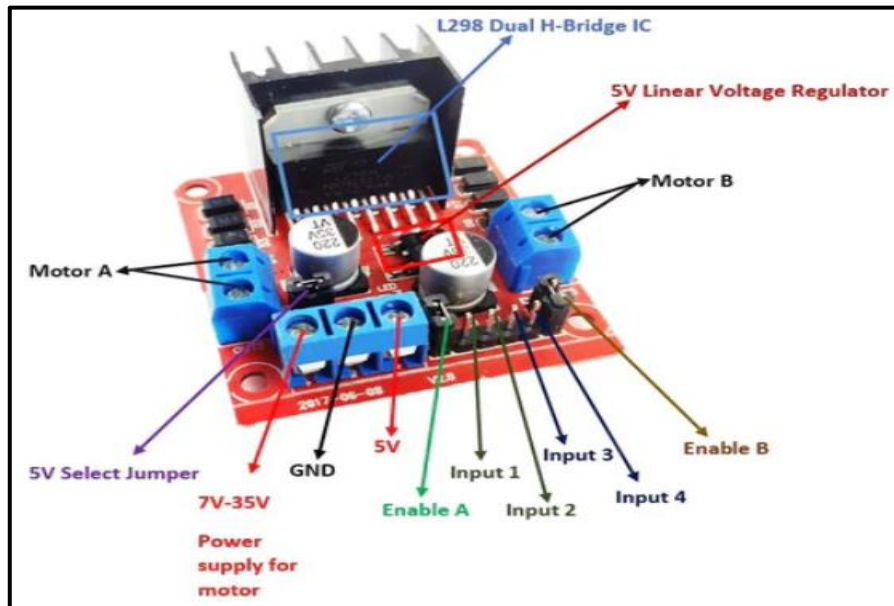


Fig:4.3 Motor Driver Module

The motor driver module (such as L293D or L298N) is used to control the direction and speed of DC motors in the robot. Since the Arduino Nano cannot provide sufficient current to drive motors directly, the motor driver acts as an interface between the controller and motors.

It enables safe and efficient operation by supplying the required current and voltage. We use this module to achieve forward, backward, left, and right movements of the robot. The main pins include **IN1, IN2, IN3, IN4** for direction control, **ENA and ENB** for speed control using PWM, and **VCC, GND** for power supply. This module ensures reliable motor operation and protects the Arduino from damage.

◆ Key Features

- Driver IC: L293D (Dual H-Bridge)
- Motor Control: 2 DC motors (or 4 motors in pairs)

- Operating Voltage: 4.5V – 36V
- Current Capacity: ~600 mA per channel
- Direction Control: Forward & Reverse
- Enable Pins: Used for speed control (PWM)

4.5 DC Motors



Fig:4.5 DC Motors

DC motors are responsible for the movement of the robot, converting electrical energy into mechanical motion. These motors drive the wheels, enabling the robot to move forward, backward, and turn in different directions. Geared DC motors are used because they provide higher torque at lower speeds, which is ideal for carrying the robot's weight and maintaining stability during operation.

They are connected to the motor driver module, which controls their speed and direction based on signals from the Arduino Nano. Due to their simple construction, reliability, and cost-effectiveness, DC geared motors are widely used in robotic applications like this floor cleaning system.

The positive terminal of dc motor is connected to batteries positive terminal and negative terminal is connected to the common ground .As motors are directly powered from batteries .

◆ Key Features

- Speed: 200 RPM (Revolutions Per Minute)
- Type: DC Geared Motor
- Operating Voltage: 6V – 12V

4.6 Ultrasonic sensor

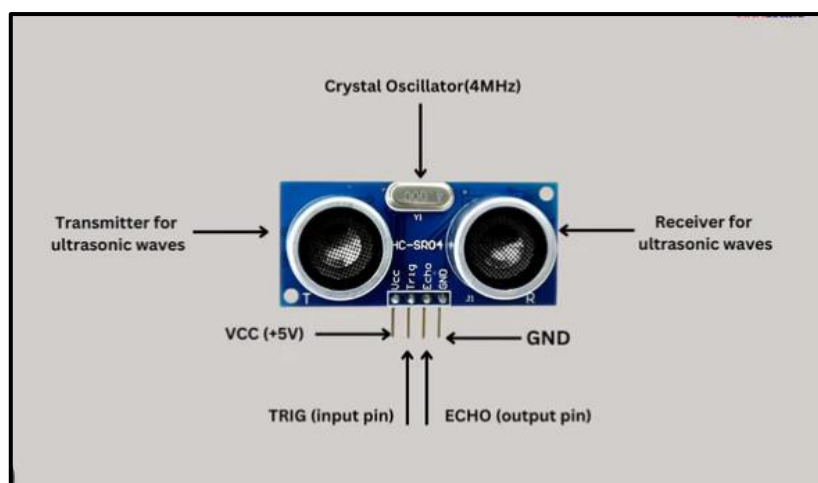


fig.4.6 Ultrasonic sensor

The ultrasonic sensor (HC-SR04) is used for obstacle detection in the robot. It works by emitting ultrasonic sound waves and measuring the time taken for the echo to return after hitting an object. Based on this time, the distance between the robot and the obstacle is calculated.

This helps the robot avoid collisions and navigate safely in automatic mode. The sensor continuously scans the surroundings and sends distance data to the Arduino Nano, which then decides the movement of the robot accordingly. Due to its accuracy, low cost, and simple interfacing, it is widely used in robotics.

Arduino sends a signal to TRIG → Sensor emits ultrasonic waves → Waves reflect from object → ECHO pin receives signal → Time is calculated → Distance is determined

◆ Key Features

- Operating Voltage: 5V
- Working Range: 2 cm to 400 cm
- Accuracy: ~3 mm
- Measuring Angle: ~15°
- Pins: VCC, GND, TRIG, ECHO

4.7 16x2 Green LCD Display



Fig:4.7 16x2 LCD Display

The **16x2 LCD Display** with an **I2C module** is used to display real-time system status such as movement direction, cleaning mode (manual/automatic), and obstacle detection alerts. It enhances user interaction by providing clear and readable visual feedback about the robot's operation.

The display works using liquid crystal technology to control light and form characters on the screen. While it normally supports both 4-bit and 8-bit modes, using an I2C module simplifies communication by reducing the required connections. Instead of multiple data pins, it uses only two communication lines: **SDA (data)** and **SCL (clock)**.

The LCD can be easily interfaced with microcontrollers like Arduino using libraries such as **LiquidCrystal_I2C**, making programming simple and efficient. Due to its low cost, low

power consumption, and ease of use, it is widely used in embedded systems and robotics projects.

◆ Key Points

- Operating Voltage: 5V
- Display Type: 16x2 (16 characters × 2 lines)
- Communication: I2C (SDA, SCL)
- Interface: Requires I2C module (backpack)
- Can display letters, numbers, and special symbols
- Low power consumption and easy to program

4.8 Water Pump



Fig.4.8 water pump

The water pump module is used to provide mopping functionality in the floor cleaning robot. It sprays or releases water onto the floor, which helps in effective cleaning when combined with a mop. The pump is controlled by the Arduino Nano through a relay module or motor driver, as it requires higher current than the Arduino can supply directly.

It is typically a small DC submersible pump, making it compact and suitable for robotic applications. The pump can be turned ON/OFF based on commands in manual mode or automatically during cleaning cycles, ensuring efficient water usage and better cleaning performance.

◆ Key Features

- Type: DC Water Pump (Submersible)
- Operating Voltage: 3V – 6V (or 6V – 12V depending on model)
- Low power consumption
- Compact and lightweight
- Continuous water flow capability

4.9 Relay Module

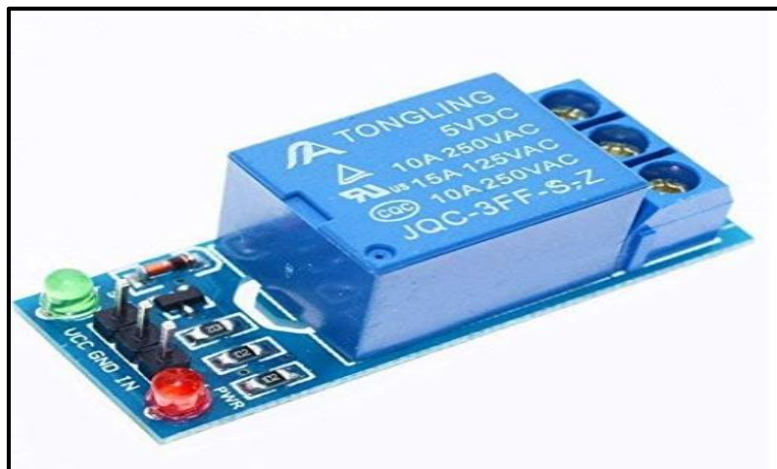


Fig:4.9 Relay Module

The relay module acts as an electrically operated switch, allowing the Arduino Nano to control high-power devices like the water pump using low-power signals. Since the Arduino cannot directly handle high voltage or current, the relay provides a safe interface by isolating the control circuit from the load circuit.

It works using an electromagnetic mechanism that opens or closes contacts when a 5V signal is applied. This makes it ideal for switching external devices ON and OFF. In the floor cleaning robot, the relay is used to control the water pump for mopping operations.

It ensures reliable switching, protects the microcontroller, and enables efficient control of higher power components.

◆ Key Points

- Operates on 5V control signal
- Terminals: COM (Common), NO (Normally Open), NC (Normally Closed)
- Provides electrical isolation between low and high voltage circuits
- Can control AC or DC loads
- LED indicator for status (in module)
- Easy interfacing with Arduino using digital pins

4.10 BreadBOARD

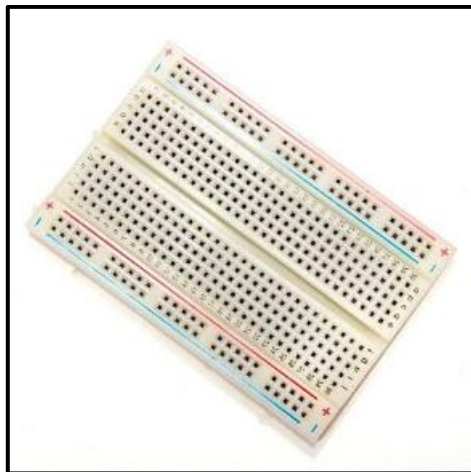


Fig:4.10 BreadBoard

The **Breadboard** is used to create temporary and flexible electrical connections in the robot during the development and testing phase. Unlike a PCB, it does not require soldering, allowing components such as the Arduino Nano, sensors, and modules to be easily inserted, removed, and rearranged.

It consists of internal metal strips that connect holes in a specific pattern, enabling quick circuit building. Breadboards are especially useful for prototyping because they allow easy modification and debugging of circuits before final implementation on a PCB.

Using a breadboard helps in testing different circuit designs, reducing errors, and saving time during development. However, it is less durable than a PCB and is mainly used for temporary setups rather than permanent installations.

◆ Key Points

- No soldering required (solderless design)
- Used for prototyping and testing circuits
- Allows easy insertion and removal of components
- Internal metal strips provide electrical connections
- Reusable and flexible for modifications
- Not suitable for permanent connections

4.11 Lithium-ion Batteries



Fig:4.11 Lithium-ion Batteries

Rechargeable lithium-ion batteries are used to provide power to the entire floor cleaning robot system. They are preferred due to their high energy density, which allows the robot to operate for longer durations without frequent recharging.

These batteries are lightweight, making them suitable for mobile robotic applications. Multiple cells can be connected in series or parallel to achieve the required voltage and current for components like motors, Arduino Nano, and water pump.

Lithium-ion batteries also have a low self-discharge rate and longer life cycle compared to other batteries. Proper charging and protection circuits are required to ensure safe operation and prevent overcharging or deep discharge.

◆ Key Points

- Voltage (per cell): 3.7V
- High energy density
- Lightweight and compact
- Long life cycle (many charge-discharge cycles)
- Low self-discharge rate
- Can be recharged multiple times
- Requires battery management/protection circuit (BMS)

4.12 Mop



Fig:4.12 Mop

The spinning mop is a key component responsible for the actual cleaning action in the floor cleaning robot. It works by rotating at a constant speed, scrubbing the floor surface to remove dust, dirt, and stains effectively. The mop is usually attached to a DC motor, which provides the required rotational motion.

When combined with water from the pump, it enhances the cleaning efficiency by loosening and wiping away stubborn dirt. The spinning action ensures uniform cleaning over the surface area, making it more effective than simple wiping.

4.14 Chassis

The chassis is the base structure that holds all components together.



Fig 4.14 Chassis

4.13 7805 Voltage Regulator

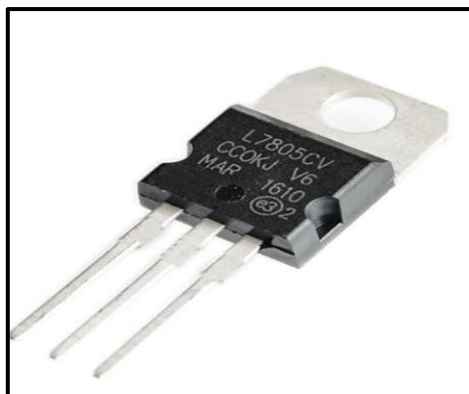


Fig:4.13 7805 Voltage Regulator

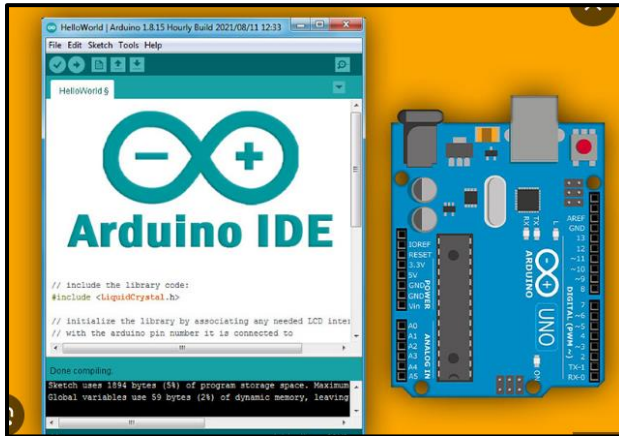
The 7805 regulator ensures a constant 5V supply to sensitive components like Arduino and sensors, protecting them from voltage fluctuations.

◆ Key Points

- Input: 7V–35V
- Output: 5V stable

5 .SOFTWARE DEVELOPMENT

5.1 Arduino ide



The Arduino IDE (Integrated Development Environment) is software used to write, compile, and upload code to Arduino boards like Arduino Nano, Uno, etc.

1. Arduino IDE :

Arduino IDE is an open-source Integrated Development Environment used for writing, compiling, and uploading code to the Arduino Uno microcontroller. It provides a simple and user-friendly interface for embedded system development. In this project, the program is written in Embedded C using Arduino IDE. The code is compiled and uploaded to the Arduino Uno using a USB cable. The IDE also provides features like serial monitoring, error detection, and debugging, which help in testing and verifying the working of the floor cleaning robot.

2. Embedded C Programming :

Embedded C is used to develop the control logic of the floor cleaning robot. It is a programming language specially designed for microcontroller-based applications. In this project, Embedded C is used to read sensor data, process Bluetooth commands, and control output devices such as DC motors, relay modules, water pump, and LCD display. The program includes conditional statements, loops, and functions to implement both manual and automatic cleaning operations effectively.

3. Arduino Libraries :

Arduino provides various built-in libraries that simplify coding and hardware interfacing. In this project, libraries such as:

- LiquidCrystal_I2C Library – used for interfacing the 16x2 LCD display
- SoftwareSerial Library – used for communication with the HC-05 Bluetooth module
- Ultrasonic sensor functions – used for obstacle detection

These libraries reduce coding complexity and help in faster development of the system.

4. Serial Monitor :

The Serial Monitor is a tool in Arduino IDE used for communication between the Arduino board and the computer. In this project, it is used to display Bluetooth commands, sensor readings, and debugging messages. This helps in testing the movement, obstacle detection, and overall performance of the robot.

5. Circuit Design and Implementation :

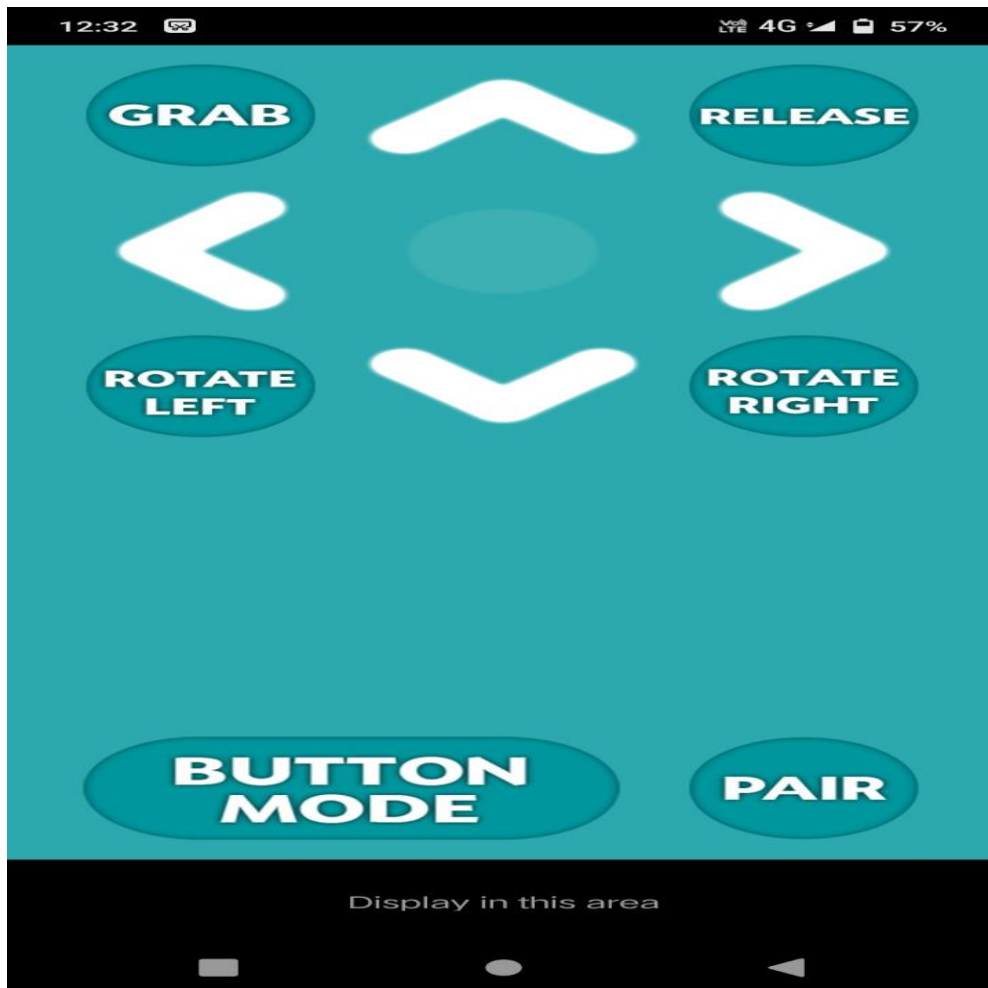
The circuit design is created using components such as Arduino Uno, HC-05 Bluetooth module, ultrasonic sensor, motor driver, relay module, LCD display, and water pump. Initially, the connections are tested on a breadboard before final implementation. Proper circuit design ensures smooth communication between hardware components and reliable robot operation.

6. Program Logic Implementation :

The core functionality of the robot is based on programmed logic. The Arduino continuously reads input data from the Bluetooth module and ultrasonic sensor. In manual mode, the robot responds to commands received from the smartphone. In automatic mode, the ultrasonic sensor detects obstacles and the robot changes direction accordingly. The cleaning mechanism and water pump are also controlled through programmed instructions.

7. Testing and Debugging :

Testing and debugging are important steps in software development. The floor cleaning robot is tested in both manual and automatic modes to ensure proper movement, obstacle detection, and cleaning operation. Sensor readings and Bluetooth commands are observed using the Serial Monitor. Errors in coding and hardware connections are identified and corrected to ensure smooth and reliable system performance.



Mobile Application Control Interface

The mobile application control interface is used to manually control the movement and operation of the floor cleaning robot through Bluetooth communication. The application is connected to the HC-05 Bluetooth module installed in the robot, allowing wireless control using a smartphone. we have used Bluestick App.

The interface consists of different control buttons such as forward, backward, left, right, rotate left, rotate right, grab, release, and pair options. These buttons send commands to the Arduino Uno through the Bluetooth module, enabling smooth and real-time movement of the robot.

The “Pair” button is used to establish the Bluetooth connection between the smartphone and the HC-05 module. Once connected, the user can control the robot easily using the directional buttons. The interface provides a simple and user-friendly method for operating the robot in manual mode.

6.COST

Sr. No.	Component	Quantity	Cost (₹)	Total (₹)
1	Arduino Uno	1	450	450
2	HC-05 Bluetooth Module	1	300	300
3	HC-SR04 Ultrasonic Sensor	3	120	360
4	DC Motors	5	150	750
5	Motor Driver Module / Relay	1	150	150
6	Lithium-ion Batteries	3	200	600
7	16x2 LCD Display	1	150	150
8	I2C Module	1	80	80
9	Water Pump	1	200	200
10	Voltage Regulator (7805)	1	20	20
11	Wheels	4	100	400
12	Chassis	1	300	300
13	Connecting Wires	1	100	100
14	PCB / Breadboard	1	150	150
15	Resistors	1 set	50	50
16	Relay Module	1	100	100
17	Male & Female Headers	1 set	50	50
18	Clamps	3	30	90
19	Nuts and Bolts	1 set	50	50
	Total	₹4200		

7.ADVANTAGES

1. Reduces human effort and physical strain.
2. Saves time in cleaning large areas.
3. Cost-effective in the long run.
4. Supports dual mode operation (Manual and Automatic).
5. Enables wireless control using the HC-05 Bluetooth module.
6. Provides obstacle detection using the HC-SR04 Ultrasonic Sensor.
7. User-friendly and easy to operate.

8.APPLICATIONS

1. Homes and residential apartments
2. Colleges and educational institutions
3. Hospitals and healthcare centers
4. Offices and corporate buildings
5. Shopping malls and commercial complexes
6. Laboratories and indoor industrial areas

9.RESULT

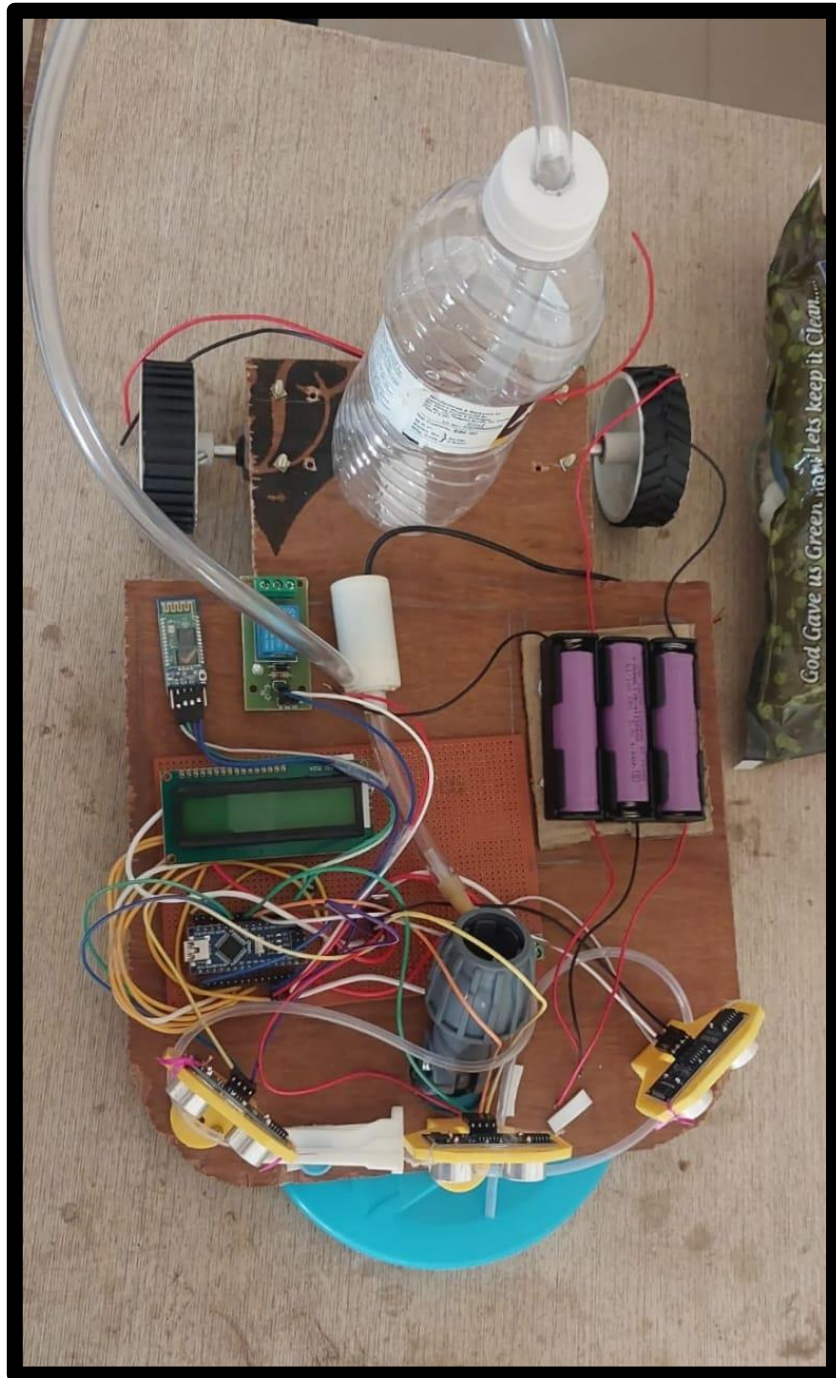
The Bluetooth controlled manual and automatic floor cleaning robot was successfully designed and implemented. The system was tested in both manual and automatic modes to evaluate its performance.

In manual mode, the robot responded accurately to commands sent from the smartphone via Bluetooth. The movement of the robot (forward, backward, left, right) was smooth and reliable. The cleaning mechanism, including the water pump and mop, functioned effectively during operation.

In automatic mode, the robot was able to navigate independently using sensors. It successfully detected obstacles and changed direction accordingly, ensuring continuous cleaning without human intervention.

The robot demonstrated good performance in cleaning small to medium floor areas. It was able to remove dust and perform basic wet cleaning efficiently. The dual-mode operation provided flexibility, allowing the user to switch between manual control and autonomous functioning as required.

First Prototype



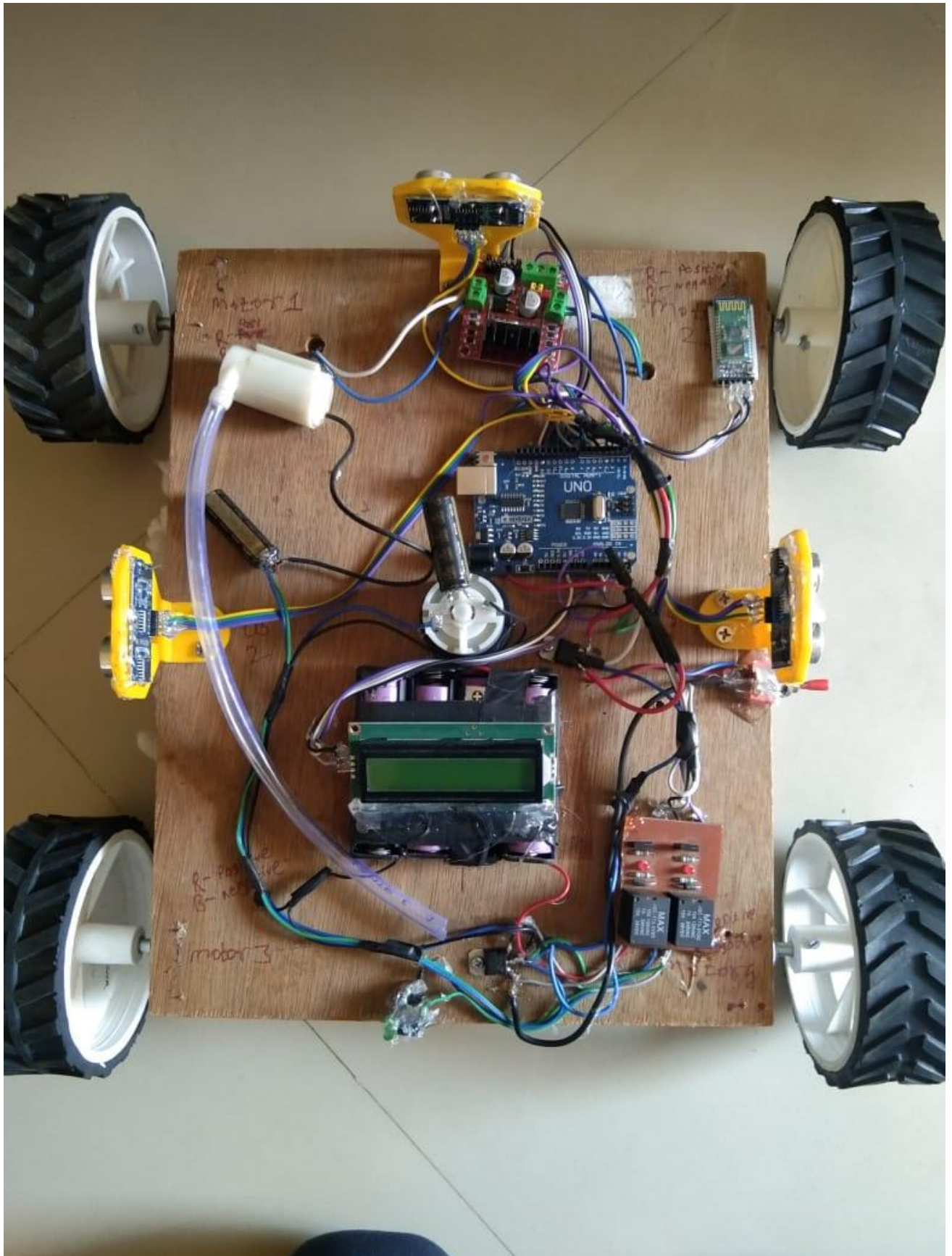
Prototype Development and Improvements

Initially, the first prototype of the floor cleaning robot was developed using a basic chassis design, breadboard connections, and a simple cleaning mechanism. After demonstrating the prototype, valuable suggestions and feedback were provided by our project guides and teachers to improve the performance, stability, and overall design of the robot.

Based on the suggestions, several modifications were implemented in the final version of the project:

- Upgraded the robot to a four-wheel drive system for better balance and movement.
- Modified the chassis structure to improve stability and component placement.
- Added a central mop mechanism for more effective floor cleaning.
- Replaced temporary breadboard connections with more reliable direct PCB/soldered connections.
- Integrated an I2C LCD module to reduce wiring complexity and minimize Arduino pin usage.
- Improved overall wiring arrangement and component positioning for better performance and appearance.

After implementing these improvements, the final version of the floor cleaning robot demonstrated better stability, smoother operation, and improved cleaning efficiency during the final project presentation.



10.CONCLUSION

The Bluetooth controlled manual and automatic floor cleaning robot was successfully designed and developed to simplify the cleaning process. The system effectively combines both manual control using Bluetooth and automatic operation using sensors, providing flexibility and ease of use.

The robot is capable of performing cleaning tasks efficiently with the help of its mopping mechanism and water dispensing system. In manual mode, the user can control the robot conveniently through a smartphone, while in automatic mode, the robot navigates independently by detecting obstacles.

The project demonstrates how automation and simple electronic components can be integrated to create a practical and cost-effective solution for everyday problems. It reduces human effort, saves time, and improves cleaning efficiency.

Overall, the proposed system meets its objectives and can be successfully used in homes, offices, and other environments for floor cleaning purposes.

11.References

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- <https://ieeexplore.ieee.org/document/10223665>
- <https://ieeexplore.ieee.org/document/10223665>
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- YouTube Tutorial Video on Bluetooth Controlled Floor Cleaning Robot Project, <https://youtu.be/ecNfo2phP6g>

. Learning Outcome

This project provided us with valuable technical knowledge and practical experience in the field of embedded systems, robotics, and automation. During the development of the Bluetooth controlled manual and automatic floor cleaning robot, we learned the complete process involved in transforming an idea into a working real-time project.

Initially, we studied the problem statement and explored different possible solutions for implementing an automated floor cleaning system. We presented our approach and project planning to our teachers and project guide, who provided valuable suggestions and approval for further development.

After the project idea was finalized, we started collecting the required components and began the hardware development process. The chassis of the robot was designed and prepared in the college workshop according to the required measurements and structure. Once the chassis was completed, individual hardware components such as the Arduino Uno, Bluetooth module, ultrasonic sensors, motor driver, LCD display, relay module, and water pump were tested separately before integration.

After assembling all components, we implemented the circuit connections and developed the software using Arduino IDE and Embedded C programming. The code was uploaded to the Arduino Uno, followed by continuous testing and debugging of the system to ensure proper robot movement, obstacle detection, Bluetooth communication, and cleaning operation.

Throughout the project development, multiple presentations and review meetings were conducted with our project guide to discuss the progress, challenges, and improvements in the system. This helped us understand the importance of project planning, teamwork, communication, and documentation in real-world engineering projects.

The project also taught us how to handle practical challenges such as component failures, circuit issues, wiring problems, debugging errors, and cost management. Overall, this mini project enhanced our technical skills, problem-solving abilities, teamwork, and practical understanding of automation systems.

This project was successfully completed as a part of the Third Year Engineering Mini Project under the Savitribai Phule Pune University curriculum with Team of Three Member's